National University of Singapore

Bridging Mathematics

Semester 1, AY2025/2026

Time allowed: 1 hour and 30 minutes

Mock Test

INSTRUCTIONS TO CANDIDATES

(a) Write down your name clearly in the space provided above.

(b) Only this booklet will be collected at the end of the examination. Do not insert any loose pages in this booklet.

(c) This paper contains **eight (8)** questions and comprises **twenty (20)** printed pages. Answer **ALL** questions to the best of your ability.

(d) This is a CLOSED BOOK examination. All calculations and working must be presented and laid out systematically.

(e) Unless otherwise stated, all numerical answers must be given exactly.

FOR EXAMINER USE ONLY

1	2	3	4	5	6	7	8

You may use this page for your rough work.

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List of Commonly Used Maclaurin Series

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \cdots \quad \text{for } |x| < 1$$

$$\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots \quad \text{for } |x| < 1$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \quad \text{for all } x \in \mathbb{R}$$

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} = \frac{x}{1!} - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots \quad \text{for all } x \in \mathbb{R}$$

$$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots \quad \text{for all } x \in \mathbb{R}$$

Question 1 [15 marks]

(a) Write down the derivative of $y = \tan x$. Hence, or otherwise, compute the derivative of the inverse tangent function $y = \tan^{-1} x$.

(b) Use integration by parts to evaluate $\displaystyle\int \tan^{-1} x \, dx$.

(c) Determine the first five terms in the Maclaurin series for $y = \tan^{-1} x$.

Show your work to Question 1 in the space below.

You may use this page to continue your work on Question 1.

Question 2 [10 marks]

Consider the indefinite integral

$$\int \frac{1}{(x-a)\,(x-b)}\,dx,$$

where a, b are some real numbers.

(a) Evaluate the integral if $a \neq b$.

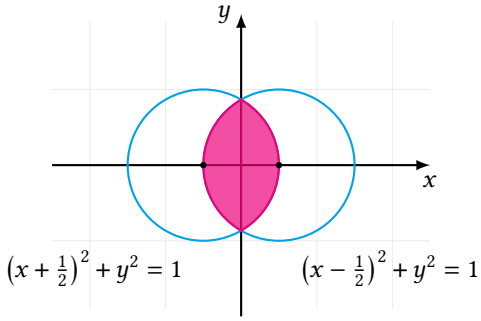
(b) Evaluate the integral if $a = b$.

Show your work to Question 2 in the space below.

You may use this page to continue your work on Question 2.

Question 3 [10 marks]

Two circles of radius 1 are made to overlap such that the center of each circle lies on the boundary of the other, as shown in the diagram below.



Determine the volume of the solid obtained by rotating the overlapping region about the x-axis.

Show your work to Question 3 in the space below.

You may use this page to continue your work on Question 3.

Question 4 [15 marks]

Determine the values of x for which the following series converges:

$$\sum_{n=1}^{\infty} \frac{1}{2^n} \tan \frac{x}{2^n}.$$

For the values of x you have found, find the corresponding sum.

Hint: You may use the fact that $\tan(x/2) = \cot(x/2) - 2\cot x$.

Show your work to Question 4 in the space below.

You may use this page to continue your work on Question 4.

Question 5 [10 marks]

(a) Write down the Maclaurin series for $f(x) = \cos\left(x^3\right)$.

(b) Hence, or otherwise, determine the value of $f^{(18)}(0)$. You may leave your answer in terms of factorials.

Show your work to Question 5 in the space below.

You may use this page to continue your work on Question 5.

Question 6 [15 marks]

Let $\vec{v}_1$ be a unit vector, and $\vec{v}_2$ be a vector of length 10 such that $\vec{v}_1 \cdot \vec{v}_2 = 5$. We define a *sequence* of vectors:

$$\vec{v}_3 = \mathrm{proj}_{\vec{v}_1}\vec{v}_2, \quad \vec{v}_4 = \mathrm{proj}_{\vec{v}_2}\vec{v}_3, \quad \ldots, \quad \vec{v}_n = \mathrm{proj}_{\vec{v}_{n-2}}\vec{v}_{n-1}.$$

(a) Compute $|\vec{v}_3|$, $|\vec{v}_4|$, and $|\vec{v}_5|$.

(b) Hence, or otherwise, show that the following infinite series converges, and compute its sum:

$$\sum_{n=1}^{\infty} |\vec{v}_n|.$$

Show your work to Question 6 in the space below.

You may use this page to continue your work on Question 6.

Question 7 [10 marks]

A pyramid in xyz-space can be described by the equation

$$|x| + |y| + |z| \leq 1, \text{ where } z \geq 0.$$

You are in a helicopter viewing the pyramid from above, directly facing a side of the pyramid given by the plane

$$\Pi : \; x + y + z = 1.$$

(a) Suppose that your helicopter is situated at the point $P\,(3, 4, 10)$. Determine your distance from Π.

(b) Find the vector equation of the line ℓ normal to Π through the point P.

(c) Determine the coordinates of the point on Π that is closest to P.

Show your work to Question 7 in the space below.

You may use this page to continue your work on Question 7.

Question 8 [15 marks]

A helical curve is parametrized by the vector function

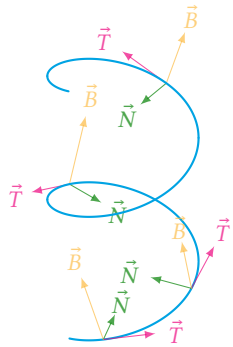
$$\vec{r}(t) = t\,\vec{i} + \cos t\,\vec{j} + \sin t\,\vec{k}, \text{ for } t \in \mathbb{R}.$$

(a) Compute the *unit tangent vector* $\vec{T}(t) = \dfrac{\vec{r}\,'(t)}{|\vec{r}\,'(t)|}$.

(b) Compute the *unit normal vector* $\vec{N}(t) = \dfrac{\vec{T}\,'(t)}{\left|\vec{T}\,'(t)\right|}$.

(c) Compute the *binormal vector* $\vec{B}(t) = \vec{T}(t) \times \vec{N}(t)$, and verify that $\vec{B}(t)$ is a unit vector.

These three vectors form an frame of reference made up of orthogonal vectors at each point on the curve $\vec{r}(t)$.



Show your work to Question 8 in the space below.

You may use this page to continue your work on Question 8.

You may use this page for your rough work.

You've made it to the end!